

# PAPER\_1587\_MU\_0\_LEAD\_1\_257

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## PAPER\_1587 — Vacuum Permeability $\mu_0 \approx \mu_0 =$ $K_{\text{MEX}} - \Phi_{5/6} + F^2 \cdot \text{SSQ} = 1.257$

**Author:** Daniel T. Murphy  
**Framework:** UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+  
**Tier:** Electromagnetism  
**Date:** June 18, 2026  
**Status:** CLOSED — 0.103% residual near-EXACT closure

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### Observation

PAPER\_1209DD\_S616 (Electromagnetism Unified Proof Set) gives the lead-digit form:

``

Vacuum Permeability  $\mu_0 \approx \mu_0 = K_{\text{MEX}} - \Phi_{5/6} + F^2 \cdot \text{SSQ} = 1.257$

``

Residual to CODATA: **0.103%**.

### UQFF Closed Identity

``

$\mu_0 = K_{\text{MEX}} - \Phi_{5/6} + F^2 \cdot \text{SSQ} \approx 1.257$  residual 0.103%

``

The expression uses only locked integer/real UQFF primitives — no fitted constants.

### Cross-Domain Pattern

Together with PAPER\_1494-1585, this paper extends UQFF's reach into atomic-scale EM and quantum-mechanical constants. The dominant integer-primitive terms across EM/Quantum constants reveal a **scale ordering**:

- **EM-coupling constants** ( $\epsilon_0$ ,  $\mu_B$ ,  $k_e$ ) cluster around  $K_{\text{MEX}} \cdot D_{\text{phys}} = 8.33$  or  $N_{\text{CH}} = 9$
- **Planck/quantum constants** ( $h$ ) cluster around  $D_{\text{BSFG}} = 6$
- **Relativistic constants** ( $c$ ) cluster around  $SO_{5/D_{\text{phys}}} = 2.5$

Different integer-primitive ranges naturally correspond to different physical scale hierarchies.

### NOT REPLACEMENT

CODATA treats Vacuum Permeability  $\mu_0$  as a precision-measured constant. UQFF supplies a structural lead-digit expression demonstrating that the integer primitives encode rational approximations to the empirical value.

### Reference

- Source: PAPER\_1209DD\_S616
- Related: PAPER\_1209DD/EE remaining catalog
- Calculator dispatch: `calculate_paradox({"paradox": "mu_0_lead_1_257"})`

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